

MITRE ATT&CK & Cyber Kill Chain

Objective: Learn the techniques and tactics threat actors use to compromise a system using the MITRE ATT&CK website.

1. What I Learned Today

I studied the seven steps of the Cyber Kill Chain and how each must be completed for a threat actor to execute a successful intrusion or attack.

The steps are:

- Reconnaissance - Information gathering
- Weaponization : Information from reconnaissance is used to develop a weapon to carry out the attack on a specific target.
- Delivery : Delivering weapon to victim through email or other means
- Exploitation: The malicious code is executed on a vulnerable system by triggering the code.
- Installation: This is when a backdoor is created on the system to maintain persistent access.
- Command and Control (CnC): This is to establish a command and control channel to communicate and remotely control the target system.
- Actions on Objective: This step describes threat actors achieving their primary objective e.g Data theft, DOS, etc.

I also studied the MITRE ATT&CK framework and why it is very important and how understanding them helps SOC analysts analyze threat actors behaviours.

2. MITRE ATT&CK Summary - 7 Tactics Every SOC Analyst Should Know

- **Reconnaissance:** This is the attacker's information gathering phase. They collect data about the target environment, exposed services, employee details, and technologies used.
- **Initial Access:** Attackers attempt to break into the environment using methods such as phishing, exploiting public-facing applications, or using stolen credentials.
Mitigation: Email security monitoring, endpoint alerts, and blocking suspicious login attempts.
- **Execution:** Malicious code is executed on a system after initial access. This includes running scripts, payloads, PowerShell commands, or malware.

Mitigation: Monitor process creation events, script execution, and unauthorized PowerShell or command-line activities.

- **Persistence:** Attackers maintain long-term access even after reboots or password changes. They may install services, scheduled tasks, backdoors, or browser extensions.

Mitigation: Watch for unusual startup items, registry modifications, and new services.

- **Privilege Escalation:** Attackers try to gain higher-level permissions (e.g., from user to admin). This allows them to control systems and move deeper into the network.

Mitigation: Detects credential dumping, token manipulation, UAC bypass, and exploitation attempts.

- **Defense Evasion:** Adversaries attempt to hide their presence and avoid detection by disabling security tools, clearing logs, obfuscating code, or using signed binaries.

Mitigation: Monitor endpoint security events, log tampering, EDR bypass attempts, and suspicious system modifications.

- **Lateral Movement:** Once privileged, attackers move across systems to reach sensitive data or high-value assets. Techniques include remote desktop, pass-the-hash, and exploiting internal systems.

Mitigation: Monitor east-west traffic, remote login attempts, and SMB/WinRM behavior.

- **Command and Control (C2):** Attackers establish communication channels to remotely control compromised systems using DNS tunneling, HTTPS, cloud services, or custom protocols.

Mitigation: Identify suspicious outbound connections, beaconing behavior, encrypted tunnels, and unusual domain patterns.

- **Exfiltration:** Attackers steal data by compressing, encrypting, or transferring it outside the network via cloud services, email, or command and control channels.

Mitigation: Identify unusual outbound connections, large data transfers, and unauthorized uploads.

3. Why These Tactics Matter for SOC Analysts

Understanding these core tactics helps analysts to:

- Map alerts to attacker behavior
- Build better detection rules
- Responds more accurately to incidents
- Improve threat-hunting and investigation depth.

4. Summary

The MITRE ATT&CK Framework and Cyber Kill chain are one of the most important knowledge bases for every SOC Analyst. It outlines the techniques attackers use at each stage of intrusion, helping analysts detect, investigate, and respond to threats more effectively.